

Study Questions: Cognition

1. Define **Cognition** in your own words, specifically as it applies to Middle School.
2. How does the Free Energy Principle constrain our understanding of Cognition?
3. Contrast the Classical view of Cognition with the Active Inference view.
4. What is a **mental model**? Give three examples of mental models you use every day without thinking about them.
5. Why do you act differently around your friends than around your grandparents? Explain using the concept of mental models.
6. Describe a mental model you had in elementary school that no longer works well in middle school. What changed?
7. What is **metacognition**? Why is thinking about your thinking a useful skill for school and for life?
8. Your brain's mental models are like a GPS map. What happens when the map is outdated? Give a real example from your life.
9. A student believes "I am bad at science." How is this a mental model? What evidence might update it?
10. Explain the three metacognition moves: **Name it, Question it, Test it**. Apply all three to a situation where you feel nervous about a presentation.
11. Why are mental models always at least a little bit wrong? Is that a problem, or is it actually useful?
12. You have a mental model of how your best friend will react to a joke. What happens when they react differently than you expected?
13. Compare a mental model to a video game strategy guide. How are they similar? How are they different?

14. Give an example of **confirmation bias** in middle school -- where someone only notices evidence that supports what they already believe.
15. A new student transfers to your school. What mental models does your brain immediately start building about them? Where does that information come from?
16. How can two students in the same class have completely different experiences? Use the concept of mental models in your answer.
17. Describe a situation where it would be smart to STOP and check your mental model before acting. What questions would you ask yourself?
18. How is debugging a mental model (finding out where it is wrong) similar to debugging a problem in real life?
19. If you could upgrade one of your mental models right now, which one would it be and why? What new information would you need?
20. Create a "Mental Model Report Card" for yourself with three models you use often. Rate each one: Is it accurate? Is it outdated? Does it need an update?